# Shivayan Chakravarty

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## Education

**B.Tech, Electronics and Telecommunication Engineering** **2023 – 2027**
*Assam Engineering College, Guwahati* *CGPA: 7.87*

**Higher Secondary Certificate** **2022**
*Sai RNS Academy* *91%*

**Senior Secondary Certificate** **2020**
*Maharishi Vidya Mandir, Guwahati* *97%*

## Technical Skills

**Languages:** Java, JavaScript, TypeScript, Python, C, C++, SQL
**Development:** Node.js, Express.js, React.js, Next.js, Tailwind CSS, REST APIs, GraphQL, Socket.IO
**Data & Tools:** MongoDB, PostgreSQL, Redis, Git, GitHub, GitHub Actions, Postman, Cloudinary
**Engineering:** Data Structures and Algorithms, Object-Oriented Programming, High-Level Design, Low-Level Design, Distributed Systems, Competitive Programming, Computer Networks, Operating Systems, DevOps

## Experience

**Software Engineer Intern** **May 2026 – Jul 2026**
*VrixaaLabs Pvt Ltd* *Hybrid*

- Engineered 3 production-oriented backend improvements in a **Node.js/TypeScript microservices platform** - centralized **RBAC authorization**, utility refactoring, and global **Redis-based rate limiting**.
- Strengthened **distributed-system infrastructure** with **Redis Pub/Sub**, Socket.IO Adapter and clustered-cron strategies, global request throttling, and **MongoDB connection-pool tuning**
- Architected a **centralized permission evaluation pipeline** aggregating memberships, entity-role assignments, and role-derived permissions; exposed the effective permission set through **GraphQL** and integrated it into entitlement authorization, eliminating fragmented permission evaluation across backend flows and improving consistency.

## Projects

**Vector - Social Media Platform** | Live | Source Code **Jan 2026**
*Next.js, Node.js, Express.js, MongoDB, Socket.IO*

- Built a full-stack social platform with **REST APIs**, cookie-based **JWT authentication**, and **Google OAuth**.
- Implemented **real-time chat** using **WebSockets** with scalable state-driven UI updates.
- Developed social features including profiles, posts, likes, comments, follows, and user interactions.

**Syntropy - AI Powered Mock Test Platform** | Live | Source Code **Aug 2025**
*Next.js, Express.js, MongoDB, TypeScript, Redis, RabbitMQ, Docker, Gemini, Groq*

- Engineered **microservices** using Node.js, TypeScript, MongoDB and **Redis caching** for frequently accessed data.
- Implemented **asynchronous workflows** with **message queues** for emails, document processing, and AI evaluation.
- Built **AI pipelines** with PDF extraction, text preprocessing, prompt optimization, and response validation workflows.
- Implemented **WebSockets** for real-time timed assessments, synchronized countdowns, and live session state updates.

## Coding Profiles

- LeetCode: 500+ problems solved
- CodeForces: Specialist (1419 rated), 150+ problems solved

## Positions of Responsibility

**Project Admin** **May 2026 - July 2026**
*Nexus Spring of Code* *Remote*

- Merged 200+ pull requests, mentored 50+ contributors, and helped grow the project to 30+ GitHub stars.

**President** **Jul 2025 – Jun 2026**
*Assam Engineering College* *Guwahati*

- Organized 15+ technical events, mentored students in DSA, competitive programming, and software engineering.